(iii) $5a-10b-4bc+2ac$
$5a-10b+2ac-4bc$
$5(a-2b)+2c(a-2b)$
$(a-2b)(5+2c)$

(iv) $x+y-1-xy$
$x-1+y-xy$
$(x-1)+y(1-x)$
$(x-1)-y(x-1)$
$(x-1)(1-y)$

$(a-b)=-(b-a)$

3.5.1 Factorisation using Identity

(i) $a^2+2ab+b^2 \equiv (a+b)^2$ (ii) $a^2-2ab+b^2 \equiv (a-b)^2$

(iii) $a^2-b^2 \equiv (a+b)(a-b)$ (iv) $a^2+b^2+c^2+2ab+2bc+2ca \equiv (a+b+c)^2$

(v) $a^3+b^3 \equiv (a+b)(a^2-ab+b^2)$ (vi) $a^3-b^3 \equiv (a-b)(a^2+ab+b^2)$

(vii) $a^3+b^3+c^3-3abc \equiv (a+b+c)(a^2+b^2+c^2-ab-bc-ca)$

Note

$(a+b)^2+(a-b)^2=2(a^2+b^2)$; $a^4-b^4=(a^2+b^2)(a+b)(a-b)$

$(a+b)^2-(a-b)^2=4ab$; $a^6-b^6=(a+b)(a-b)(a^2-ab+b^2)(a^2+ab+b^2)$

Progress Check

Prove: (i) $\left(a+\frac{1}{a}\right)^2+\left(a-\frac{1}{a}\right)^2=2\left(a^2+\frac{1}{a^2}\right)$ (ii) $\left(a+\frac{1}{a}\right)^2-\left(a-\frac{1}{a}\right)^2=4$

Example 3.25

Factorise the following:

(i) $9x^2+12xy+4y^2$ (ii) $25a^2-10a+1$ (iii) $36m^2-49n^2$

(iv) x^3-x (v) x^4-16 (vi) $x^2+4y^2+9z^2-4xy+12yz-6xz$

Solution

(i) $9x^2+12xy+4y^2 = (3x)^2+2(3x)(2y)+(2y)^2 \left[\because a^2+2ab+b^2=(a+b)^2\right]$
$= (3x+2y)^2$

(ii) $25a^2-10a+1 = (5a)^2-2(5a)(1)+1^2$
$= (5a-1)^2 \quad \left[\because a^2-2ab+b^2=(a-b)^2\right]$

(iii) $36m^2-49n^2. = (6m)^2-(7n)^2$
$= (6m+7n)(6m-7n) \quad \left[\because a^2-b^2=(a+b)(a-b)\right]$

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